

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12} . The ratio of their molar solubilities i.e. $\frac{S(A_2X)}{S(MX)} = \underline{\hspace{2cm}}$.

(Round off to the Nearest Integer).

Q2: 16 March (Shift 2) - Numerical

Sulphurous acid (H_2SO_3) has $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$. The pH of 0.588M H_2SO_3 is (Round off to the Nearest Integer)

Q3: 17 March (Shift 1) - Numerical

0.01 moles of a weak acid HA ($K_a = 2.0 \times 10^{-6}$) is dissolved in 1.0 L of 0.1M HCl solution. The degree of dissociation of HA is $\times 10^{-5}$
(Round off to the Nearest Integer).

[Neglect volume change on adding HA . Assume degree of dissociation $\ll 1$]

Q4: 18 March (Shift 1) - Numerical

In order to prepare a buffer solution of pH 5.74 sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is M. (Round off to the Nearest Integer).

[Given: pK_a (acetic acid) = 4.74]

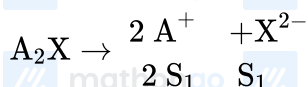
Q5: 18 March (Shift 2) - Numerical

The solubility of $CdSO_4$ in water is $8.0 \times 10^{-4} \text{ mol L}^{-1}$. Its solubility in 0.01M H_2SO_4 solution is
(Round off to the Nearest integer) (Assume that solubility is much less than 0.01M)

Answer Key**Q1 (50)****Q2 (1)****Q3 (2)****Q4 (10)****Q5 (64)**

Q1 (50)

For A_2X



$$K_{sp} = 4S_1^3 = 4 \times 10^{-12}$$

$$S_1 = 10^{-4}$$

for MX



$$S_2$$

$$K_{sp} = S_2^2 = 4 \times 10^{-12}$$

$$S_2 = 2 \times 10^{-6}$$

$$\text{So } \frac{S_{A_2X}}{S_{MX}} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$$

Q2 (1)

H_2SO_3 [Dibasic acid] $c = 0.588M$

$\Rightarrow$ pH of solution p due to First dissociation only since $K_{a1} \gg K_{a2}$

$\Rightarrow$ First dissociation of H_2SO_3



$$t = 0 \quad C$$

$$t \quad C - x \quad x \quad x$$

$$\Rightarrow K_{a1} = \frac{1.7}{100} = \frac{[H^+][HSO_3^-]}{[H_2SO_3]}$$

$$\Rightarrow \frac{1.7}{100} = \frac{x^2}{(0.58-x)}$$

$$\Rightarrow 1.7 \times 0.588 - 1.7x = 100x^2$$

$$\Rightarrow 100x^2 + 1.7x - 1 = 0$$

$$\Rightarrow [H^+] = x = \frac{-1.7 + \sqrt{(1.7)^2 + 4 \times 100 \times 1}}{2 \times 100} = 0.09186$$

Therefore pH of sol. is : $pH = -\log[H^+]$

$$\Rightarrow pH = -\log(0.09186) = 1.036 \simeq 1$$

Q3 (2)

Hints and Solutions

MathonGo



Initial conc. 0.01M 0.1M 0

Equ. conc. $(0.01 - x)(0.1 + x)x$ M

$$\approx 0.01\text{M} \approx 0.1\text{M}$$

$$\text{Now, } K_a = \frac{[\text{x}^+][\text{A}^-]}{[\text{HA}]} \Rightarrow 2 \times 10^{-6} = \frac{0.1 \times x}{0.01}$$

$$\therefore x = 2 \times 10^{-7}$$

$$\text{Now, } \alpha = \frac{x}{0.01} = \frac{2 \times 10^{-7}}{0.01} = 2 \times 10^{-5}$$

Q4 (10)

$$\text{pH} = \text{pK}_a + \log \frac{[\text{CB}]}{[\text{WA}]}$$

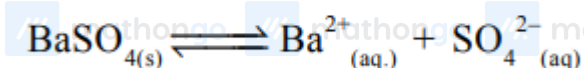
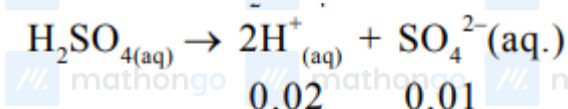
$$5.74 = 4.74 + \log \frac{[\text{CB}]}{1}$$

$$\Rightarrow [\text{CB}] = 10\text{M}$$

Q5 (64)

$$\text{In pure water, } K_{sp} = S^2 = (8 \times 10^{-4})^2$$

$$= 64 \times 10^{-8} \text{ In } 0.01\text{M H}_2\text{SO}_4$$



$$K_{sp} = x(x + 0.01)$$

$$K_{sp} = x(x + 0.01)$$

$$= 64 \times 10^{-8}$$

$$x + 0.01 \cong 0.01\text{M}$$

$$\text{So, } x(0.01) = 64 \times 10^{-8}$$

$$x = 64 \times 10^{-6}\text{M}$$